

SUBMITTED ARTICLE



Dynamic changes in the structure and concentration of the international grain and oilseed trading industry

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Abstract

This paper aims to describe the evolution of firm and industry strategies and to analyze detailed transaction data on the structure of the international grain trading industry. Specifically, data are developed from vessel nominations, and these were used to derive measures of concentration and determine how these change across selected geographies and commodities, as well as to identify the clusters of firms in this industry. The results provide unique documentation on the structure and concentration of the world grain trading industry. The results indicate the levels of concentration are much less than traditionally suggested; there has been growth by many new entrants making up a viable competitive fringe; the industry seems to be comprised of three clusters of firms; and generally, competition in this industry would be characterized as being “fierce.”

KEYWORDS

concentration, grain trading, industry structure

JEL CLASSIFICATION

Q13, Q17, A18

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The international grain trading industry has undergone significant and rapid changes over the past five decades. It has transitioned with an increased number of firms and evolving government interventions. Recent years have witnessed notable shifts in the industry's structure, the emergence of a viable competitive fringe, including the rise of trading firms in China and Russia, and extensive geopolitical interventions resulting in numerous and varying trade restrictions. Despite these policy changes, there has been a dearth of in-depth analysis regarding the industry's structure. Measures of concentration have evolved, yet they often overlook the influence of new market entrants, a substantial competitive fringe, and the impact of Russian and Chinese trading firms.

This paper aims to describe the evolution of firm and industry strategies and to analyze detailed transaction data on the structure of the international grain trading industry. Earlier studies have analyzed concentration in international grain trading, generally indicating it is highly concentrated and dominated by four firms commonly called ABCD (ADM, Bunge, Cargill and Dreyfus). However, these studies are older (the most recent used data from the early 2000s), even though there have been many changes in the past decades, use aggregated data, ignore new entrants and the competitive fringe, and do not account for the growth of Chinese and Russian grain trading firms. In this study, we utilize data on vessel nominations to (1) derive measures of concentration, (2) assess how these measures vary across different geographical locations and commodities, and (3) determine the composition of strategic groups, or clusters of firms, within the industry.

The results of this study offer insights into the structure and concentration of the global trading industry. This study makes four key contributions. First, it utilizes a novel dataset previously inaccessible within this industry. The data encompass export sales for individual shipments, which are unique and serve to derive concentration measures over time and among countries and firms. Second, the analysis demonstrates the effects of the growing competitive fringe and the rise of trading entities in China and Russia, which are influencing the industry's structure. Third, the findings reveal that concentration levels are significantly lower than those reported in prior studies. Finally, the results describe the composition of firms within three clusters, providing a more accurate characterization of this industry.

The first section below summarizes previous studies, highlighting changes in the structure and strategies of firms within the international grain trading industry and the growth of Chinese and Russian grain trading firms. Subsequent sections present the data sources, methods, and results. Two concentration measures are derived for total shipments of grain and oilseeds from and to major exporters and importers. The final section offers a summary and implications.

BACKGROUND AND PREVIOUS STUDIES

Firms in the international grain trading industry have been impacted by numerous evolving institutional, policy, and geopolitical changes.¹ These include the evolution from surpluses and the introduction of export subsidy programs and biofuels, the demise of many State Trading Enterprises (STEs) in international trading, the escalation in volatility (Bullock et al., 2023; Franke et al., 2023; Rechtsteiner et al., 2023) and implications for risk management, the increase in government interventions following the Ukraine invasion (Reidy, 2024). Olam's Chief Executive indicated the "world is headed for 'food wars' as geopolitical tensions were pushing countries into conflicts" (Speed & Savage, 2024). Finally, the growth in exports from Russia and Ukraine following the 1990s and growth in imports by China in the 2000s impacted



the competition structure, as discussed below. Russia and Ukraine's exports of corn, soybean, wheat, oils and meals increased from 40 to 119 mmt between 2012 and 2024. From 2012 to 2023, Chinese imports of the same commodities increased from 61 to 137 mmt (Fastmarkets, 2025). Hence, there was substantial growth in both cases.

The apparent strategies that dominated/prevailed by firms in this industry have also evolved. Before the 1980s, the trading environment was generally characterized as a few largely private firms; most information (e.g., prices [basis], export sales, and shipments) was not public; export firms were mainly not vertically integrated. Caves (1977/1978) provided an early strategic interpretation of the international grain trading industry. A few firms controlled the international grain market, commonly known as ABCD.² This structure was partly due to two sources of economies of scale: economies of scale with facilities (storage and handling), shipping, and economies of scale for information, which was intangible and asymmetric.

Changes began to emerge in the 1990s, highlighting the importance of optionality (Johansen & Wilson, 2018; Meersman et al., 2012; and as explained by Blas & Farchy, 2021; Kingsman, 2021). Optionality refers to the ability to shift origins and terms of trade, thereby inducing multiple origination capabilities in response to volatility and/or Black Swan events. In volatile markets, there is value in switching origins, which provides incentives for firms to develop origination capabilities from multiple origins and ultimately favors larger firms. Finally, there was a structural change because information became more readily accessible. This includes information regarding crop conditions, weather, prices, basis, shipping costs, ship loading, wait times, tenders, and tender results. Before the advent of digital dissemination, these data were only partially available to non-principals in transactions and were typically lagged.³ Hence, the traditional source of asymmetrical information advantage was lessened with near-ubiquitous information owing to digital technology (Belt & Boudier, 2020; Belt & Porsborg-Smith, 2020; Reidy, 2021). This situation resulted in a shift of functions from “having” information to “data analytics” (Vanian, 2018). Thus, given firms have virtually equal and near-instantaneous access to information, most export firms sought to escalate their data analysis capabilities, which would be integrated with trading.⁴

In the early 1990s, grain firms' strategies evolved toward greater vertical integration, and grain trading firms expanded into flour milling, malting, oilseed crushing, biofuels, and varying livestock sectors. More recently, most trading firms are extending their functions to manage sustainability and carbon initiatives. Commencing in the 1990s, rail shipping had two significant changes. One was the change from single-car shipping to multicar shipping.⁵ The second was developing a tradable market for rail cars in the United States, which was adopted in other countries (Wilson & Klebe, 2024; Wilson & Lakkakula, 2021). Concurrently, there was a shift involving most grain trading firms to become supply chain managers (Bertelsen, 2018; Ching & Lau, 2012).

There are at least two important structural and strategic implications of these changes. One is that optional-origin (or multi-origin) exporting as a trading strategy whereby trading firms became “masters of optionality” which has escalated in importance. The second is for an increase in new entrants, ultimately comprising growth in the competitive fringe in this industry.

Mergers, new entrants, and growth of competitive fringe

The major firms in the modern international grain trading industry were Andre, Bunge, Cargill, Continental, Dreyfus, and Cook. Each was a large private and family-controlled firm. Over

time, some of these exited, others were acquired or became insolvent, and the remaining ABCDs were subject to public ownership or financial disclosure in varying means (e.g., as public stock-held firms, ESOPs, etc.). Since the 1990s, there have been numerous mergers, including of significance: Cargill acquired the assets of Continental in 1999, ADM had numerous acquisitions in the 2000s, Glencore expanded in grain in 2002 and acquired Viterra in 2012, LDC had several regional acquisitions in 2013 (InVivo & Soufflet, 2021; Marubeni & Gavilon, 2013; Viterra & Gavilon, 2022). The most recent is Bunge's acquisition of Viterra, which commenced in 2023 and has been reviewed by most of the major grain producing and importing countries, and was approved by the EU in August 2024, and Canada in January 2025 (Government of Canada, 2025) subject to conditions. The outstanding review (as of early 2025) is that from China.

In addition, numerous firms entered the international grain trading industry, in addition to the new Chinese trading organization and Russian grain trading firms (discussed below). These new entrants and others ultimately form the competitive fringe of firms in international grain trading. Without being exhaustive, the new entrants can be classified into several categories, including cooperatives (e.g., CHS, Zen-Noh), STEs, and private enterprises. Japanese trading companies (non-STEs) include Marubeni, Mitsubishi, Itochu, and Korean trading firms Pesco, Daewoo, and CJ International Asia. In addition, Zen-Noh and Itochu own CGB (Consolidated Grain and Barge). Other noteworthy entrants include Noble, Olam, and Wilmar, referred to as the *Asian Tigers* or *Merchants of the Orient* (Wang & Li, 2010). In 2025, SALIC acquired the remaining shares of Olam Agri (Okoh On, 2025). Private commodity trading companies entered, including Phibro, Ferruzzi, Gavilon, Agrex (Mitsubishi), Glencore, J. Aron, BGT, ETG, Serentz, Atlas, Greenfield, Hartree Partners (ex-Goldman / HessGunvar), and Vitol. Others entered, including Grain Corp (Australia), Lansing Grain (now Andersons), Quadra (Australian Wheat Board [AWB] traders in Geneva), InVivo (who now owns Soufflet), and Copenhagen Merchants. Following the Russian invasion of Ukraine and the subsequent exit of some major players from Ukraine trading, several new Ukrainian firms expanded into trading and some later exited. Concurrently, there was an increase in the number of Russian grain trading firms.

These de novo and, in many cases, greenfield expansions into international grain trading indicate that entry is not prohibitive.⁶ Some were successful (e.g., Gavilon, Viterra, Glencore, and others); others were not and either exited or were acquired. In most cases, these firms realized (1) that trading and managing logistics of international grain trades are more complicated than envisioned (i.e., as compared to trading futures); (2) the importance of counterparty risk; and (3) that multiorigin optionality is important, each of which favored larger firms. Two significant additions to the above relate to the entry and/or expansion of trading for Chinese imports and Russian exports.

Chinese trading

China emerged as a large and growing market for international agricultural commodities commencing in about 2000. China became the dominant soybean importer, and more recently, it became a dominant importer of corn, wheat, and beef cattle. As a result of this growth, China sought to diversify its suppliers and sources and “wanted less dependence on ABCD” (Roberts, 2014). Ultimately, it was said that “The Chinese Want Their Own Cargill” (Roberts, 2014), and more recent policies pursued diversification (National Development and Reform Commission, 2023). COFCO International grew to assume the responsibility for

managing a portion of China's imports.⁷ They initially acquired Nidera, Noble AGRI LLC, and subsequently others.

COFCO International was formed in 2014 and is a subsidiary of COFCO, a state-owned enterprise (COFCO International, [n.d.](#)). In 2024, they had operations in 37 countries. COFCO has heavily invested in supply chain functions, including origination, storage, ports, and shipping, and has logistical capacities in South America, Eastern Europe, and the United States. COFCO International continues to grow. In mid-2024, COFCO swapped assets with Growmark to access the US Gulf ports, indicating that it intends to expand at both the US Gulf and PNW (Clayton, 2024).⁸ COFCO ships to China and is a significant shipper to many other countries. It is also a large shipper of grains from the United States and other countries. COFCO has grown in revenues and has been planning an IPO (since 2021). COFCO International agribusinesses now have revenues second to Cargill in 2022, exceeding those of ADM, Viterro, Wilmar, Bunge, and other major agribusiness firms.⁹ Taken together, COFCO International has many characteristics of an STE.¹⁰

Russian grain trading

The Russian grain trading industry has evolved radically and is essential in world trade. Traditionally (pre-1990s), Russia's grain trade was controlled by Exportkhleb (Crawford, 2022). Under Perestroika, the industry was largely decentralized (Wilson & Belozertsev, 1995), and various forms of commodity markets have evolved. Major international grain trading firms expanded into varying functions within the interior and offshore markets but by no means dominated the industry.

Russian trading firms also evolved. VTB (<https://www.vtb.ru>) sought help from the Kremlin to create a Russian grain champion to curb the role of foreign traders (Houghton, 2019). Russia became concerned about food security when Western sanctions were imposed in 2014. In 2019, VTB consolidated its role in local grain marketing and expanded into trading, logistics, and port handling. VTB's grain holding company, Demetra, intended to control the supply chain and become a multinational giant; the company was partly owned by private firms and partially state-owned through VTB. Concurrently, United Grain Co. (<https://ozk-group.ru>) became a commercial company, was primarily state-owned, and sought to control the supply chain (it owns facilities, rail cars, etc.), becoming a primary exporter from the Black Sea. Taken together, Russia evolved with two competing firms: state-owned and quasi-state-owned.

In late 2022, the Kremlin issued a decree prohibiting companies of “persons related to unfriendly states” from buying grain from Russian farmers. This action reduced trading opportunities for non-Russian firms and increased profits for Russian-trading firms. In early 2023, these developments, among others, effectively forced Western agricultural trading firms (including Cargill, LDC, and Viterro, as well as an earlier autonomous exit by Bunge) to liquidate their assets and exit Russia's grain-trade sector (Popva & Plume, 2023; Terazono, 2023). The structure of the exporting grain industry in Russia has evolved radically (Glauber, 2023; IFPRI, 2023; Quinn, 2024a, 2024b). Early on, Grain Gates was the dominant exporting firm, followed by TD RIF (Grain Flower, previously named GTS) and Aston. Quinn (2024a) indicated that the top five exporters shipped 58% of the exports. The only multinationals reported were Dreyfus and COFCO. In the 2023/2024 crop year report, Quinn (2024a) reported that the Russian grain trading firms increased their market share. By 2023/2024, most of the Western trading firms had exited. Grain Gates, TD RIF, and Aston were the most prominent private grain exporting firms.

TD RIF has since exited (AgriCensus, 2024; Quinn, 2024b), and Grain Gates is formally a private company but associated with Demetra, which, as a private company, is associated with VTB.

Taken together, the critical points for the purpose of this study are that (1) Russia is a significant exporter, particularly of wheat; (2) following privatization, many western trading firms were active in Russia's grain exports; (3) due in part to sanctions in 2014, there were efforts to grow domestic Russian grain trading firms; and (4) following the Ukraine invasion, western firms were forced to exit, replaced by a cabal of Russian trading firms which are now consolidating to a few exporters.

Evolution of concentration measures in international grain trading

Concentration in the international grain trading industry has been the subject of numerous earlier studies and books and, more recently, referenced in more popular studies and media. These are summarized in Table 1. Academic literature and industry references on concentration in international grain trading have evolved and are described below. These results vary widely in several responses. Some studies use international shipments, US export shipments, sales, or

TABLE 1 Summary of studies indicating CR4's for international grain trading.

Author	Year of publication	Scope	Year of study	All grains	Corn	Soybean	Wheat
Caves	1977	International		80			
		US domestic	1960	33			
			1972	21			
Caves and PÜgel	1982	US exports	1976		42	41	61
Thompson and Dahl	1979		1976	85			
McCalla and Schmitz	1979		1970	90 (CR5)			
Folz	2002	US exports	1999		81	65	47
Crespi and MacDonald	2022	US Exports	1998				47
			2009				65
Other books and Studies							
Frievalds	1976	International		70			
		US exports		90			
Morgan	1979			80–90			
Gilmore	1982	US exports			90		96
Atkin	1992	International		75			
Kingsman	2021	International		50			
Murphy	2012	International		73			

Note: Morgan's estimates were for the United States, EU, Canada, and Argentina and range from 80 to 90. For "Other books and studies," the study date is generally the date of publication or not specified.

domestic shares. Most, but not all, of the studies use physical quantities. Not all of the studies were clear on these metrics.

The early academic study was conducted by Caves (1977/1978), who indicated a concentration ratio of 80% in international grain trading, although sources were not identified. Caves (1977, p. 109) indicated the CR4 for US domestic grain merchants at 33% and 21% in 1960 and 1972, respectively. Caves and Pugel (1982) suggested that for 1974/1975, an unofficial CR4 of 61%, 42%, and 41% for wheat, corn, and soybean, respectively. CR4's fell for US exports to 1980/1981. Using data from NAEGA, the 1974 CR4 for wheat, corn, and soybean was 62%, 43%, and 53%, respectively. Thompson and Dahl (1979) indicated a CR4 of 85% and questioned the ability to coordinate pricing dynamically. McCalla and Schmitz (1979) indicated a CR5 of 90% in 1970. Using data from USDA Federal Grain Inspection Service (FGIS), Foltz et. al. (2002) indicated concentration ratios for US grain exports of 81%, 47%, and 65% for corn, wheat, and soybeans respectively. These studies are not all directly comparable, but they suggest relatively high concentration, and there has not been a noticeable change over time.

Several books allude to concentration in international grain trading. Freivalds (1976, p. 116) indicated a CR5 of 90% for the United States and 70% for world grain trading. Morgan (1979) indicated that the CR5 by origin country was 85%, 90%, 90%, and 80% for the United States, EU, Canada, and Argentina, respectively. He referred to the “pyramid of power” and the dominance of the prominent grain exporting firms (pp. 234–235). Gilmore (1982) reported values for US exports of wheat and corn at 96% and 90%, respectively. Sewell (1992, p. 127) indicated the concentration “...is largely due to these risks and pressures that the international grain trade became concentrated into so few hands...” Atkin (1992, p. 112) suggested that the five major grain trading firms account for 75% of international grain shipments but acknowledged that the exact figure is unknown. More recently, Kingsman (2021, p. 222) indicated that ABCD controlled 50% of the world trade in grain and oilseeds.

Other more recent publications have cited varying concentration measures in international grain trading. These measures, mainly from Murphy et al. (2012), indicate a CR4 of 73%. The basis of that study was an AWB reference.¹¹ Upon further investigation, it became clear that the study, which addressed privatizing the Australian grain trading industry, had significant limitations. The analysis summed the total revenues from public annual reports for Cargill, ADM, ConAgra, Louis Dreyfus, Bunge, CWB, and AWB and derived the share controlled by the largest firms. It was inferred from these data that these firms control 73% of the market share for international grain trading. However, this approach is limited partly because it is based on a priori firm composition and of firms having publicly accessible data. Further, using revenues as a proxy for market size in international grain trading would severely underestimate the market size, ignoring actual shipments and the impacts of competitive fringe. Specifically, the study only used data from the selected public firms. As a result, it ignores revenues, or more importantly, volumes, from the rest of the industry. Thus, the industry's total size is underestimated, creating an upside bias to their CR4 estimate.

Other publications describe concentration in international grain trading (Anderson et al., 2023; Clapp, 2015; Harvey, 2022; Hietland, 2024; Murphy et al., 2012) indicating that concentration in international grain trading ranges from 70% to 90%. These references are important because they have been the subject of numerous stories in the financial trade press seeking to describe concentration in intentional grain trading (Farmer, 2017; Kokkinidis, 2023; Lawrence, 2011; Pina, 2017; Putz, 2018; Thomas, 2023). Each of these either refers to Murphy et al. (2012) or Clapp (2015) or does not provide a source, but each indicated the concentration in international grain trading is 70%–90%. Most of these studies claim that concentration has

escalated, which has had adverse consequences. Murphy et al. (2012) indicated, “these four firms are decisive actors in the global restructure of the overlapping food, feed, and fuel complexities.” Clapp (2015) indicated that “these firms have enormous power to shape key aspects of the global food landscape.” Public Eye (2019) suggests a connection to human rights violations. ETC Group suggested these firms “wield enormous influence over markets, agricultural research, and policy development, which undermines food sovereignty.”

Hietland (2024) suggested that these firms control the supply side through their large storage capacity, vertical integration, and so forth, and control the key export and import markets. Another indicated: “The fact that global commodity giants are making record profits at a time when hunger is rising is unjust and is a terrible indictment of our food systems” (Share the World's Resources, 2022). Anderson et al. (2023) suggested that “big ag” was sufficiently large to influence regulations. Finally, Lianos et al. (2024) described concentration in the broader agricultural sectors (e.g., fertilizer, grain, seeds, etc.) and identified many potentially adverse consequences of further concentration. In the case of international grain trading, they referred that “the ABCD companies ... hold substantial market shares, which heightens the risks associated with food security” (p. 41), and refer to the 73% CR4 as referenced above. No new evidence about concentration in this industry was provided.

DATA SOURCES ON INTERNATIONAL GRAIN TRADING

The data used in this study are observations from individual shipments (nominations) of grains and oilseeds. Thus, each observation records the quantity shipped for an individual shipment. The data were taken from various sources that are generally publicly accessible, in some cases by subscription. These sources included international shipping agencies, charter companies, and so forth.^{12,13} The commodities included are corn, soybean, wheat, and individual vegetable oils. Data for the individual shipments were classified as free-on-board (FOB) or cost and freight (CNF).¹⁴ FOB shipments are for grain sales loaded on a ship at the origin, and the importer manages and accrues the shipment cost and risk. CNF shipments are for sales in which the importer manages and accrues the cost and risk of shipping to the import port.

There is an important distinction between the data used in this study and earlier studies. The data reflect the exporter who sold the grain and arranged shipment or provided the grain to the buyer who arranged the shipment. Some earlier studies used FGIS data representing the firm that loaded the ship. Data from these studies may or may not be the firm that made the export sale. In most countries, it is not uncommon for the loader to load a shipment for another exporter that makes the export transactions and manages the execution of the sale (e.g., Cargill loads a vessel sold to COFCO).¹⁵

Some shipments were where the shipper or receiver was “unspecified.” These are relatively minor: 21% of the FOB and 22% of the CNF shipments were unspecified. Upon further inspection, in most cases, there was a more significant share of unidentified for CNF versus FOB shipments. There was a greater concentration of “unspecified” shipments in the EU, Black Sea, and Danube origins in 2023, probably due to shippers wanting to mask their identities. For these shipments, the only source of information was from the inspection agency records (SGS,¹⁶ which does not provide buyer and seller information), and these did not provide information on the buyer or seller.¹⁷ In addition, many shipments from small trading firms represented flows for cross-border shipments from Russia to neighboring countries and to border cities in

China (e.g., Heihe City). These were included in the overall data but, as described below, did not impact the derivation of the concentration measures.

In summary, the data comprised 62,271 FOB shipments and 48,098 CNF shipments. To verify that the data were representative, we derived the portion of world trade reflected in the data. The data comprise about 80% of world trade and the oilseeds market and is substantial for the major commodities (corn, soybean, and wheat). This dataset is highly disaggregated and is far less aggregated than the specification of market shares used in previous studies (as described above).

RESULTS

CR4s and Herfindahl–Hirschman index

The four-firm concentration ratio (CR4) was the sum of the market shares of the four largest firms. Specifically, CR4 was defined as:

$$CR4 = \sum_{i=1}^4 s_i,$$

and the Herfindahl–Hirschman index (HHI) was defined as:

$$HHI = \sum_{i=1}^n s_i^2 \cdot 10,000,$$

where n is the number of firms in the industry, and s_i is the i th firm's market share (ranked from largest to smallest) expressed in decimal format (i.e., 5% would be 0.05). The CR4's reported here included unspecified shipments in the total market size (the denominator). This implicitly assumes that the unspecified firms are not included among the largest four firms. For HHI, any export firm with less than 1% market share was excluded, assuming implicitly that unspecified firms are from this category. This does not materially affect the derivation of HHI and is an accepted practice in defining concentration (Besanko et al., 2016, p. 287). The HHI produces a value that ranges from near zero (extremely competitive industry) to 10,000 (a monopoly). Generally, lower values suggest more competition, no single firm dominating, little market power, and little to no differentiation (Besanko et al., 2016).¹⁸

Table 2 summarizes the results, and Figures 1–4 illustrate the market shares of each firm from 2020 to 2023. These figures display only FOB shipments (with differences for CNF shipments outlined), covering all origins and destinations for the specified commodities, including China imports and Russian exports. Due to space constraints, only the top 20 firms are represented in the figures.

Generally, the market share of the four largest firms is around 30%, though this varies by commodity and country. The aggregate CR4 for FOB and CNF shipments is 32% and 27%, respectively. These averaged 33% during the first 3 years but fell to 27% in 2023. In all cases except for US exports, FOB shipments are less concentrated than CNF shipments. The four firms with the largest market share for FOB shipments are Cargill, COFCO, ADM, and LDC,

TABLE 2 Summary of International Grain Trading Measures of Concentration (2020–2023).

Commodity scope		FOB/ CNF	CR4	HHI	Top 4 firms
All grains/oilseeds	World	FOB	32	492	Cargill, COFCO, ADM, LDC
	World	CNF	27	442	Cofco, Bunge, ADM, Cargill
All grains/oilseeds	US Exports	FOB	45	1241	ADM, COFCO, Cargill, Zennoh
		CNF	37	670	ADM, COFCO, Bunge, Cargill
All grains/oilseeds	North Africa	FOB	26	387	Cargill, ADM, Bunge, COFCO
		CNF	22	415	ADM, Bunge, Cargill, COFCO
All grains/oilseeds	China Imports	FOB	44	855	Cargill, Cofco, ADM, LDC
		CNF	39	769	Cofco, ADM, Cargill LDC
All grains/oilseeds	Russian Exports	FOB	33	623	Trade House RIF, Aston, Grain Gates/ Mirogroup, United Grain
		CNF	26	408	Solaris, Aston, GTCS, Vitterra
Corn	World	FOB	43	723	COFCO, Cargill, ADM, BUNGE
		CNF	36	540	COFCO, ADM, Cargill, BUNGE
Soybean	World	FOB	45	930	Cargill, ADM, LDC, Bunge
		CNF	38	674	ADM, BUNGE, Cargill, LDC
Wheat	World	FOB	21	343	Viterra, Cargill, COFCO, TradehouseRIF
		CNF	16	315	CBH, Viterra, COFCO, Cargill

while the leaders for CNF shipments are COFCO, Bunge, ADM, and Cargill. Therefore, there is a distinction in market leaders between FOB and CNF shipments.

Figure 2 illustrates the market shares for shipments to North Africa, included due to its being one of the largest and fastest-growing markets. Egypt, often one of the biggest importers, is included in this region. The market has a CR4 and HHI for FOB imports of 26% and 387, and CNF shipments of 22% and 415, respectively. Several notable changes occurred in this import region. The CR4 decreased from 30% to 27% between 2020 and 2023, while the CNF CR4 dropped from 26% to 19%. Furthermore, the largest Russian trading firm in this market, Solaris, improved from 11th to 5th largest between 2020 and 2023.

The outcomes differ for China's imports, where CR4s are higher at 44% and 39% for FOB and CNF shipments, respectively. Cargill and COFCO lead in FOB shipments, while COFCO and ADM dominate CNF shipments.

There have been drastic changes in Russian exports in recent years, resulting in Western firms exiting and Russian trading firms becoming more important (as discussed above). The largest firms for FOB shipments are Trade House RIF, Aston, Grain Gates/Mirogroup, and United.¹⁹ For CNF shipments, the largest firms are Solaris, Aston, GTCS, and Viterra. The market structure for Russian exports continues to change with further consolidation of exporters. When evaluated for individual years, the CR4 increased from 33% to 44% for FOB shipments in 2020 and 2023, respectively. Trade House RIF was the largest firm each year, and Cargill fell from the fourth to the eighth largest firm. CR4s for CNF shipments range from 29%–35% with

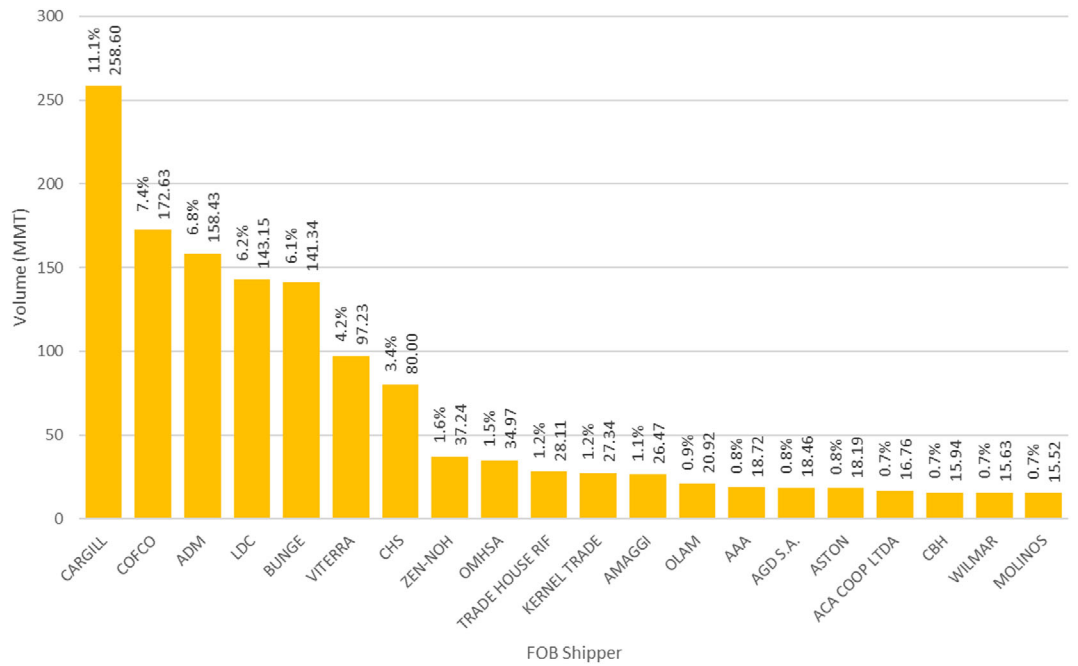


FIGURE 1 Volumes and market shares for free-on-board (FOB) shipments, top 20 firms, all commodities, all origins, and destinations, 2020–2023 (values for each bar are market share and volume).

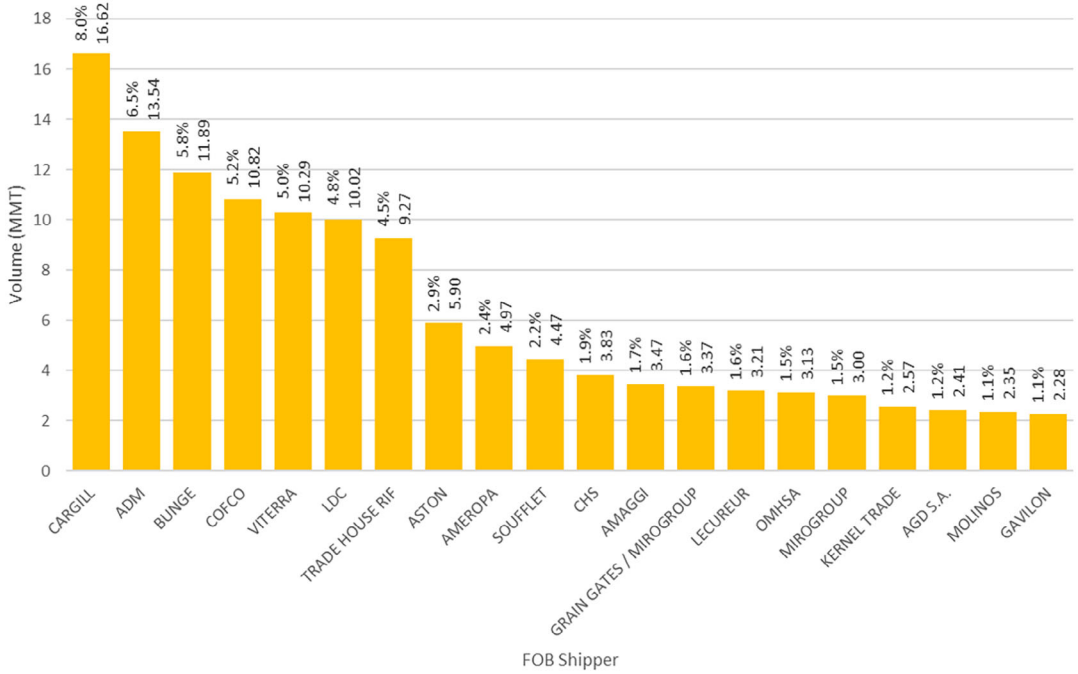


FIGURE 2 Volumes and market shares for free-on-board (FOB) shipments to North Africa, top 20 firms, all commodities, all origins, 2020–2023 (values for each bar are market share and volume).

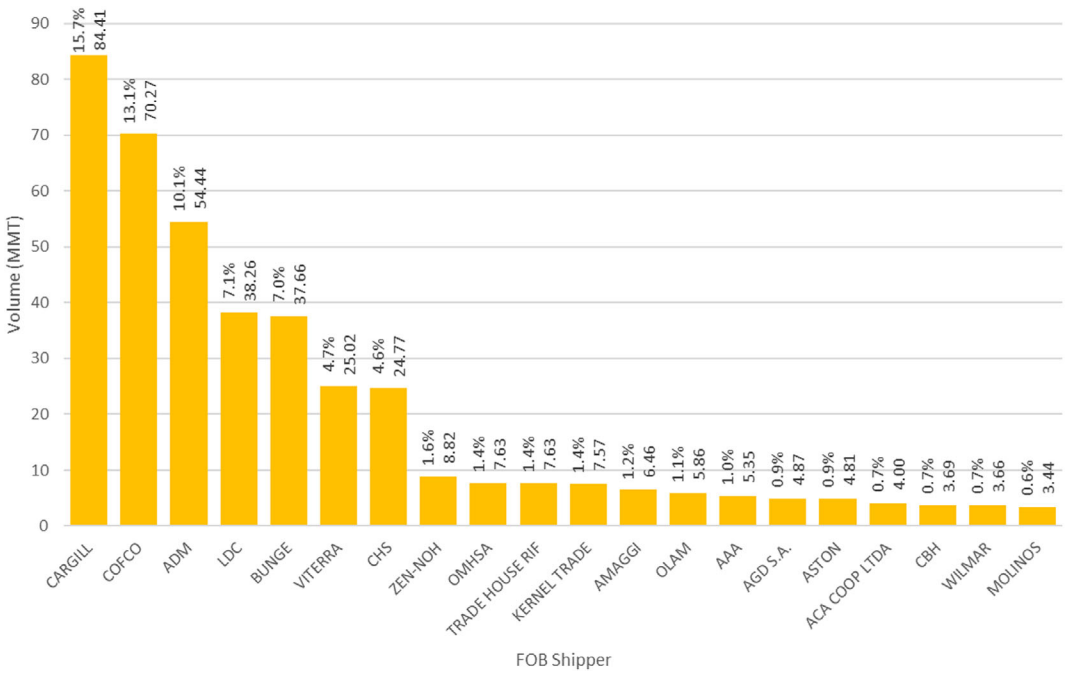


FIGURE 3 Volumes and market shares for free-on-board (FOB) shipments to China, top 20 firms, all commodities, all origins, 2020–2023 (values for each bar are market share and volume).

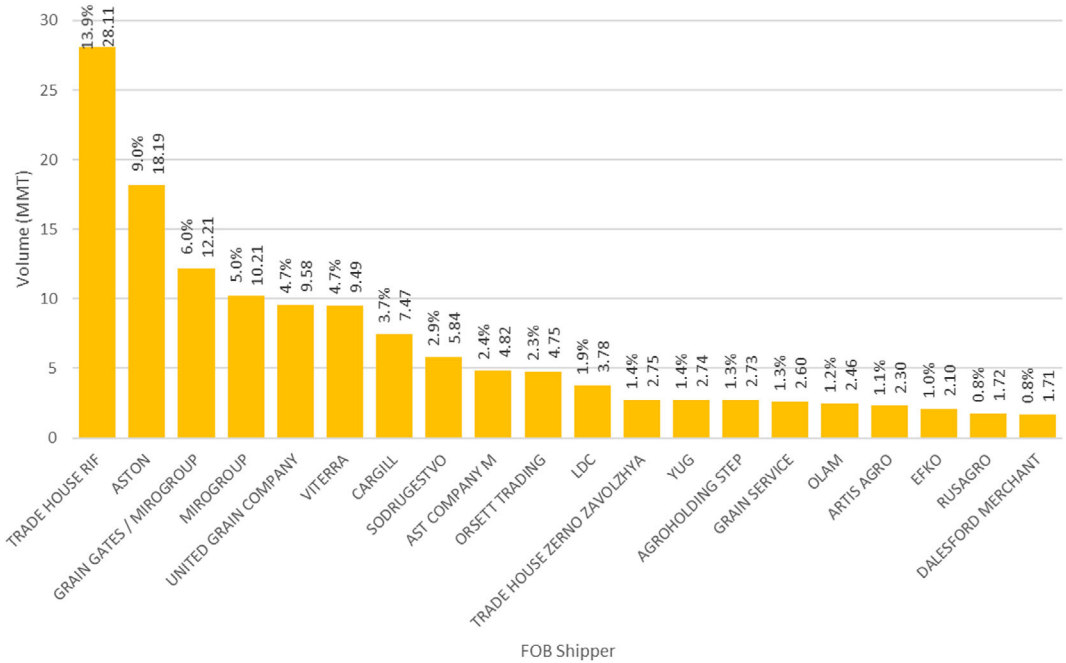


FIGURE 4 Volumes and market shares for free-on-board (FOB) shipments from Russia, top 20 firms, all commodities, all destinations, 2020–2023 (values for each bar are market share and volume).

no apparent trend. However, the largest firms shifted from 2020 to 2023, with Aston, GTCS, Solaris, and Grainflower being the largest firms in each year, respectively. Cargill became much less important (at the eighth largest).

Taken together, there have been some notable changes through time. Those of particular importance include (1) the growing penetration of COFCO for CNF and FOB China shipments and decreases by Cargill and ADM; (2) radical changes in Russia and reduction in western firms, particularly for wheat; and (3) the increase in non-Majors.

In general, these results indicate that the international grain trading industry is highly competitive. Or, these results suggest the intensity of price competition would be “fierce” (Besanko et al., 2016). There are numerous prospective reasons for this, including: (1) entry seems relatively easy or accommodative; (2) large, sophisticated and well-informed buyers with low switching costs; (3) limited product differentiation, among others. Indeed, the competitive fringe of this market has grown and is much greater than previously depicted. Based on the value of HHI, there is no reason to expect that any firm would be able to exert market power or, taken together, operate as an oligopoly.

The concentration levels reported here are significantly lower than those in previous studies, which suggested an aggregate CR4 of 73%. CR4s vary considerably across countries and commodities, and the firms identified as largest differ. Several factors contribute to this difference. First is the period: our analysis covers the most recent years and is summarized from 2020 to 2023. The latest publications include unpublished studies from the early 2000s. Much has changed in this industry during this time, particularly with the rise of competitive fringe players, including Chinese and Russian trading firms. Second is the scope of analysis. Earlier studies relied solely on data from the United States, and the AWB (2004) study, which served as a foundation for others, utilized revenue data that are now outdated and is limited as discussed above. This study considers all origins and destinations, reporting an aggregate of grain commodities as shown in Table 2. Third is the metric used. Earlier studies employed export data from the 1970s to the 1980s for shipments from the United States. The other recent study inappropriately inferred market shares from the revenues of the presumed leading grain trading firms. In this study, we utilize actual export shipments, which offer a more precise measure of trade and concentration. Finally, we derived CR4s in this study as they are commonly used to infer market structure and are comparable to previous studies that derived CR4s. In addition, we derived HHI's, which is a more suitable measure of concentration and potential market power.

Cluster analysis

The international grain trading industry has evolved to be called ABCD, implying an industry dominated by these four firms. Another interpretation is that a single cluster comprises these four firms. Cluster techniques can be used to more formally and statistically determine the composition of clusters, or more commonly referred to as “strategic groups” within an industry. Cluster analysis was used in this study to analyze the structure of the international grain industry and to more formally determine the number of clusters and the composition of firms within each cluster. The k -means clustering model assumes that the data can be partitioned into a predefined number of clusters, k .

The variables used to delineate clusters included: (1) the total number of shipments, (2) the number of origins, (3) the number of destinations, (4) the number of FOB shipments, (5) the number of CNF shipments, and (6) whether the firm trades wheat, corn, and soybeans. K -

means clustering was employed to segment firms into clusters that reveal patterns of strategic positioning. Additional variables were also tested, such as whether the firm trades in small grains, the number of shipments over time, geographic factors like continents or dependence on specific trade routes, and the average shipment size by volume. However, these additional variables were found to be insignificant and were omitted from the final analysis.

The number of clusters was determined using the Silhouette score. The Silhouette score assesses the quality of clusters produced by clustering algorithms. It measures how similar an object is to its cluster (cohesion) compared to other clusters (separation). For each point, the silhouette score ranges from -1 to 1 , where a higher score signifies better-defined clusters. A score close to 1 indicates that the point is well matched to its cluster and poorly matched to neighboring clusters. In contrast, a score near -1 suggests misclassification. By averaging these scores, the silhouette score aids in identifying the optimal number of clusters and validating clustering outcomes. The k -means algorithm then iteratively assigns each firm to a cluster by minimizing the variance within clusters, effectively grouping firms with similar trade characteristics. The Python library Scikit-learn was employed in this paper with its k -means clustering algorithm (Pedregosa et al., 2011).

The results are summarized in Table 3, with the Silhouette scores depicted in Figure 5. The Silhouette scores reveal the presence of three clusters of firms in the international grain trading industry. Cluster 1 consists of seven firms, contrasting with the traditional taxonomy of ABCD. In addition to ABCD, it also includes COFCO, Viterra, and CHS (ABCCCDV). These firms account for 45% of global FOB shipments, much lower than the earlier studies that derived

TABLE 3 Cluster results for international grain trading firms.

Cluster	1	2	3
<i>n</i>	7	37	15
Silhouette score	0.60	0.55	0.64
Firms	Cargill	Olam	Agroholding Step
	ADM	Amagg	Prometey
	Cofco	Aston	Profit
	Bunge	Sierentz	Kernel
	Viterra	Ameropa	Grain Gates
	LDC	Gavilon	Solagro
	CHS	Nibulon	Grain Service
		NCH	Enerfo
		CBH	Marubeni
		28 Other firms	6 Other firms
Feature Variables	Size of total shipments		
	Number of origins		
	Number of destinations		
	Number of FOB shipments		
	Number of CNF shipments		
	Does the company transport corn, soybean, and wheat?		

Abbreviations: CNF, cost and freight; FOB, free-on-board.

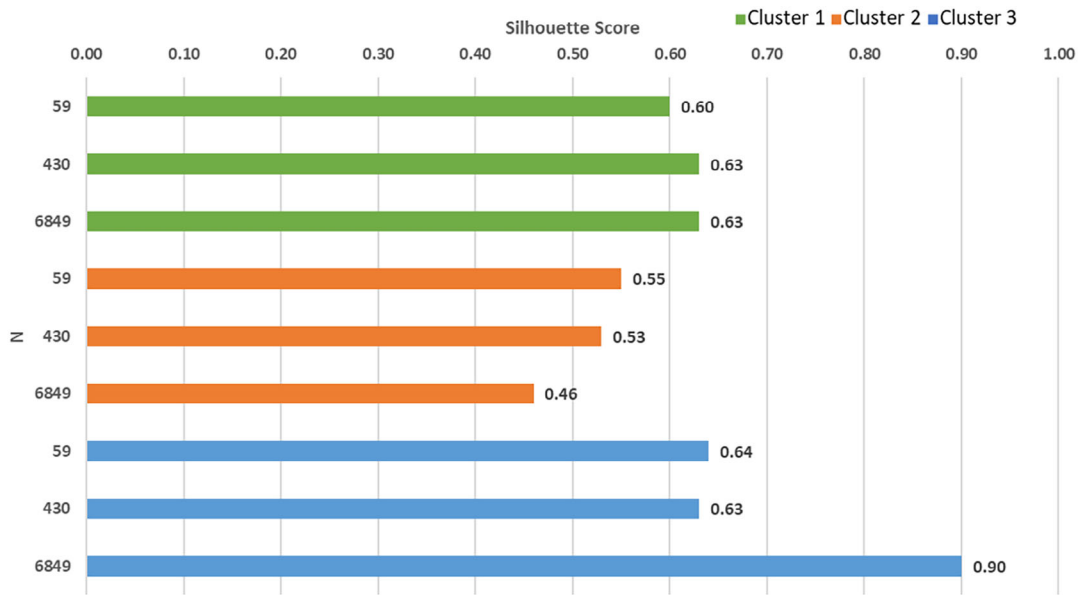


FIGURE 5 Silhouette scores for cluster evaluation.

CR4s for ABCD only. This finding is fundamental in understanding the structure of the international grain trading industry. Cluster 2 comprises nine firms with similar characteristics and many other firms, which could be seen as the competitive fringe, a broad segment of similar firms. Finally, Cluster 3 contains many other firms, each individually small.

Cluster 1 consists of firms that manage a high volume of shipments. On average, shipments from Cluster 1 firms are 67 times those of Cluster 2 firms and 107 times those of Cluster 3 firms. Cluster 1 firms, on average, ship from 14.4 origins to 92 destinations, encompassing all commodities. This aligns with the strategic concept that firm size and trading across multiple origins and destinations are crucial in this industry. Cluster 2, in contrast, has much smaller shipment volumes, with these firms shipping from an average of 1.8 origins to 11.9 destinations. Similarly, Cluster 3 differs because firms have limited origins and do not ship wheat, corn, or soybeans. Their shipments are primarily vegetable oils. Firms across all strategic groups ship both FOB and CNF, with the number of FOB shipments consistently exceeding that of CNF shipments. The significant difference between Clusters 1 and 2 underscores the importance of size and being multi-origin and multi-destination shippers.

These results are fundamental to understanding the structure of the international grain trading industry. First, rather than suggesting one segment called ABCD, there is a cluster comprising seven firms with similar characteristics. Second, a large group of other firms differ but have similar structural characteristics. These firms would be the competitive fringe. Finally, there is a large cluster (Cluster 3) with many small firms that do not handle wheat, corn, or soybeans. Taken together, this can be interpreted as an industry comprised of clusters, one of which represents larger firms that ship from a large number of origins to a large number of destinations. The competitive fringe is more complicated, as there are two clusters with many firms. The impact of the competitive fringe has not been included in earlier studies. However, as noted here, the competitive fringe is important and impacts the structure and conduct of the international grain trading industry.

CONCLUSIONS AND IMPLICATIONS

International grain trading is an old industry that has been undergoing radical changes. Significant changes in agriculture and policies, factors impacting economies of scale, and an evolving international market environment have affected it. Firm strategies have changed partly in response to these changes, and as a result of these changes, the market structure of this industry has changed.

This study aims to analyze detailed data on the structure of the international grain trading industry and document the evolution of firm and industry strategies. It used detailed data on individual export shipments for the major grains and oilseeds. Two measures of concentration were evaluated: CR4 and HHI. The latter is the most relevant measure of concentration and captures the effect of firms' number and size distribution.

The results indicated that CR4s were 27% and 32% for all commodities from all origins to all destinations. The HHI's were 885 and 442 for FOB and CNF shipments, respectively. The result varied across origins, destinations, and commodities. The CR4s for Chinese imports were 44% and 39% for FOB and CNF Chinese imports. COFCO was the largest exporter for CNF and Cargill for FOB shipments. In the case of Russia, the most prominent firms were all Russian, with Western firms being near inconsequential competitors. Finally, there was a slightly greater concentration in export trading for corn and soybeans and much less for wheat. The latter is no doubt due to the changes in Russia, as well as those in Australia and Canada.

The largest firms also differ from earlier studies. For the entire dataset, the top four firms for FOB shipment are Cargill, COFCO, ADM, and LDC, whereas for CNF, the top firms are COFCO, Bunge, ADM, and Cargill. Hence, this illustrates that the conventional representation of ABCD is incorrect. Instead, COFCO has become more prominent. COFCO is one of the largest firms for virtually every sub-aggregation measured, except notably Russia.

These results differ significantly from previously reported studies, which generally indicate CR4s in 70%–90%, or occasionally, 73%. There are several important reasons for these differences, including (1) time period, (2) scope of data, and (3) metric. Our analysis utilizes more detailed data not available before for actual shipments, and we derived both the CR4 and HHI. The results suggest that the international grain trading industry is strategically (structurally) defined as competitive and price competition may be “fierce.” In other words, the competition is so intense that the ability of any individual firm to exert a notable influence on price or trade terms is minimal.

The data underwent further analysis to identify the clusters or strategic groups within the international grain trading industry. The results revealed three clusters. Cluster 1 includes seven firms that are very large in shipment volume and transport all commodities from numerous origins and destinations. The other two clusters contain a greater number of firms, and within each cluster, the size of shipments is smaller, with firms shipping from a limited number of origins to a few destinations. Cluster 1 comprises the major players, but firms in this sector consist of more than just the ABCDs. Cluster 2 represents many smaller firms that make up the competitive fringe. Lastly, Cluster 3 is a specialized group that focuses on vegetable oils.

The findings of this study carry significant implications for both private and public sectors. For private enterprises, the results delineate alternative strategic approaches, the emergence of new market entrants, and the competitive landscape. The low HHI values suggest a highly competitive environment. The critical nature of information and the enhanced necessity for analytical capabilities have escalated due to the increased accessibility of data. Moreover, the data utilized in this study have emerged as a crucial component of information previously excluded

from strategic analyses conducted by trading firms. Lastly, it is evident that large corporations possess a level of optionality, allowing them the flexibility to transition among alternative sourcing options. Two pivotal policy issues have been identified in the preceding discussion. These findings indicate that concentration within this industry is low, according to conventional concentration metrics and compared to other agricultural sectors (MacDonald et al., 2023). Furthermore, it is imperative to recognize that the growth of grain trading organizations from China and Russia has significantly influenced the industry's structure.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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ENDNOTES

- ¹ These institutional and policy changes, as well as the evolution of strategies (discussed below) by trading firms are described in great detail in Wilson et al. (2025).
- ² This refers to the firms Archer Daniels Midland (“A”), Bunge (“B”), Cargill (“C”), and Louis Dreyfus (“D”).
- ³ As examples: Crawford (2022) described in detail the advantage Cargill and Continental had due to their internal analysis of crop conditions in negotiating the famous Russia grain deal in the 1970s; and Kingsman (2025) described how the trading industry gained competitive advantage by using local weather information to get ahead of yield and price projections.
- ⁴ Belt and Boudier, 2020 explain the impact of digitization on commodity trading and the new sources of competitive advantage. One of importance relates to the digital information which “enable companies to access and digest a growing mountain of increasingly complex data. Players must invest to increase their access to data and data-processing capabilities.”
- ⁵ Technically, multi-car shipments began in the 1970s but were not routine. The Staggers Rail Act (1980) encouraged more innovation in pricing and service. During the 1980s’ US railroads experimented with various forms of multi-car shipments. Following experiments in 1993, a more standardized form of shipment evolved which is now commonly called ‘shuttle trains. A shuttle train is a contractual arrangement facilitating a shipment to/from approved facilities, for multiple forward periods, within a shipping window each month, and the instrument is tradable which allowed development of a secondary market.
- ⁶ These are in addition to several firms seeking to digitally integrate commodity supply chains (trading with greater use of digital transactions through the supply chain). These firms include Indigo, Farmer’s Edge, FBN, GroIntelligence among others. Ultimately, these firms failed, and they exited the sector (Chipman & Hirtzer, 2024).

- ⁷ In addition to COFCO International, there are other emerging and, in some cases, large Chinese trading firms in grain, including Wilmar, XMXYG, Pengdu Agriculture, Beidahuang Group and Shenzhen Cereals Holdings Group. SinoGrain was relegated a role for storage in 2022 (Chang, 2022).
- ⁸ COFCO acquired a grain facility on the Mississippi River and Growmark received a Chicago grain warehouse facility.
- ⁹ ETC Group's Food Barons report of 2022 (Shand et al., 2022) put COFCO as the second largest company of the 'Leading Agricultural Commodity Traders'. https://www.etcgroup.org/files/files/food-barons-2022-full_sectors-final_16_sept.pdf.
- ¹⁰ Technically, COFCO International is a subsidiary of COFCO which is a STE.
- ¹¹ Murphy et al. (2012) cite the Australian Wheat Board (n.d.), which we subsequently identified that the source of this information was a non-public report referred as AWB (2004, p. 7) though conducted by Boston Consulting Group (2004).
- ¹² A detailed list of the subscriptions services and agencies are described in Appendix S1 and Wilson et al. (2025).
- ¹³ Information on commodity flows and shipping is becoming more readily accessible through the growth of varying global vessel tracking and lineup services. Most are now likely in the public domain. Indeed, this is now becoming an important element of trading firms' data analytics.
- ¹⁴ Technically, the data indicate the "charterer" but, we cannot discern whether it is a CNF or CIF sale. Hence, we refer to these interchangeably. Slabotsky (1984) defines these more formally.
- ¹⁵ It is not uncommon for export firms to use varying forms and "put-through" or "buy-sell" arrangements to facilitate loading for another export firm.
- ¹⁶ SGS (Société Générale de Surveillance SA) is a Switzerland based export inspection company.
- ¹⁷ Further detail about unspecified shipments, as well as other characteristics by commodity and for CNF shipments as well as by region and time, are provided in Wilson et al. (2025).
- ¹⁸ Interpretation of HHI relied on the economics of strategy references (e.g., Besanko et al., 2016, Chapter 5). This reference seeks an interpretation about prospective conduct and performance but must be supplemented with 'circumstances surrounding competitor interaction.' i.e., the intensity of competition depends on many other factors. Besanko et al. (2016) indicate that ranges often times attributed to HHI values are only suggestive, due to that 'generally' there are "many factors besides market structure that contribute to conduct and performance."
- The HHI is also used in merger guidelines. In the US FTC 2023 Merger Guidelines indicated markets with an HHI greater than 1800 are highly concentrated (whereas the 2010 guidelines referred to markets with HHI greater than 2500 as highly concentrated, and HHI between 1500 and 2500 as moderately concentrated). In Canada, a post-merger HHI threshold is 1800. In anti-trust enforcement, the HHI is used as a threshold and may/should be supplemented by other metrics and considerations.
- ¹⁹ These results differ somewhat from Quinn (2024b) due to that here we use data from 2020 to 2023, whereas Quinn (2024b) reports shares for 2024 and excludes vegetable oils and oilseeds. In addition, these results are through 2023, and there were additional changes in 2024 which are described in Belikova (2024).

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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